Academic Year 2026-27
FY MCA- SEM I
Python
Class Assignment

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Q1. Write a program to read a file and count how many vowels are present in its content

```
4   f = open("text.txt", "r")
5
6   data = f.read()
7   count = 0
8
9   for ch in data:
10      if ch.lower() in "aeiou":
11          count = count + 1
12
13  print("Total vowels:", count)
14
15  f.close()
```

Output:

```
PS E:\IMCC\Python\Assignment_1> py Q1.py
Total vowels: 128
PS E:\IMCC\Python\Assignment_1>
```

Q2. Read all lines from a file into a list. Print the total number of lines and display the 2nd and 4th line.

```
f = open("text.txt", "r")

lines = f.readlines()

print("Total number of lines:", len(lines))
print("2nd line:", lines[0], end="")

f.close()
```

Output:

```
PS E:\IMCC\Python\Assignment_1> py Q1.py
Total vowels: 128
PS E:\IMCC\Python\Assignment_1> py Q2.py
Total number of lines: 6
2nd line: The old clock on the mantelpiece
PS E:\IMCC\Python\Assignment_1>
```

Q3 Insert node at a specific position

```
5   sentence = input("Enter a sentence: ")
6
7   words = sentence.split()
8
9   longest = max(words, key=len)
10
11  print("Longest word:", longest)
```

Output:

```
PS E:\IMCC\Python\Assignment_1> py Q3.py
Enter a sentence: It contains a subject and a verb and can stand alone as a complete sentence.
Longest word: sentence.
PS E:\IMCC\Python\Assignment_1>
```

Q4: Find Middle node and print its value

Read a file, split its content into words, and store them in a list
i) Print the total number of words
ii)Print all unique words

```
5   file = open("text.txt", "r")
6
7   content = file.read()
8   words = content.split()
9
10  print("Total number of words:", len(words))
11
12  unique_words = set(words)
13  print("Unique words:", unique_words)
14
15  file.close()
```

Output:

```
PS E:\IMCC\Python\Assignment_1> py Q4.py
Total number of words: 12
Unique words: {'text', 'and', 'typesetting', 'dummy', 'the', 'industry', 'Lorem', 'is', 'Ipsum', 'printing', 'of', 'simply'}
PS E:\IMCC\Python\Assignment_1>
```